

Technical Requirments

BoostK

BoostK Team

10th November, 2025

Features

1. Authentication
 - a. Login / Logout for agents
 - b. Role-based access (Admin, Agent, Viewer)
 - c. Basic Profile management (name, email, department, profile photo)
2. Dashboard
 - a. Overview of:
 - i. Active chats count
 - ii. Average response time
 - iii. List of active conversations with preview and timestamps
 - b. Search and filter chats (by agent, customer, department, or keyword)
 - c. Sorting by recent or unread
3. Chat
 - a. View active chats and message history
 - b. Real-time messaging between agent and customer
 - c. Message status indicators (sent, delivered, read)
 - d. Attachments (optional: images, documents)
 - e. Transfer chat to another agent or department
 - f. Tag chats by category (e.g., Sales, Support)
 - g. End chat/archive conversation
4. Quick Responses
 - a. Predefined canned messages (editable by admin)
 - i. "Thanks for reaching out! How can I help you today?"
 - ii. "Could you provide more details?"
 - iii. "Let me transfer you to the right department."
 - b. One-click insertion into chat
 - c. Categorized quick replies (sales, support, general)
5. Upload media/files
6. Customer Management
 - a. Basic customer info panel (name, email, department)
 - b. View the customer's previous chat history
 - c. Tag or add internal notes per customer
7. Notifications
 - a. New message notifications (sound + visual badge)
 - b. Incoming chat alerts

- c. Chat transfer or assignment notifications
- 8. AI-Powered Features
 - a. Multilingual Support: Translate chats in real time
- 9. UI / UX Requirements
 - a. Two main views:
 - i. Agent Dashboard: For managing chats
 - ii. Customer View: For user-facing chat widget
 - iii. Clean, modern design (similar to Intercom / Crisp / Zendesk Chat)
 - iv. Quick response buttons below the chat area
 - v. Responsive layout (desktop, tablet)
 - b. Message bubbles with timestamp and sender distinction
 - c. Typing indicator for both sides
 - d. File upload button with progress indicator and preview
- 10. Real-Time & Backend Requirements
 - a. Realtime Chat: WebSockets / Firebase Realtime
 - b. Database: PostgreSQL
 - c. Authentication: JWT or AWS Cognito
 - d. File Storage: AWS S3
 - e. API: RESTful or GraphQL
 - f. Notifications: Web push or Firebase Cloud Messaging
 - g. Deployment: Docker-ready backend with Nginx reverse proxy
- 11. Tech Stack (Web App)
 - a. Frontend: React + Vite
 - b. State Management: Zustand / Redux Toolkit
 - c. UI Library: Tailwind CSS + ShadCN UI
 - d. Real-time Messaging: Socket.io
 - e. Backend: Node.js (NextJS)
 - f. Database: PostgreSQL
 - g. Hosting: AWS

Image for reference:

